

Arush Jain

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Summary

B.Tech student with a strong analytical and quantitative foundation, experienced in building data-driven systems that quantify financial impact and surface operational discrepancies before they become costly — inventory cost-optimization models, experiment audits that uncover hidden revenue risk, and SQL reporting pipelines that turn raw transactional data into stakeholder-ready financial insights. Comfortable building reliable software tools with AI-assisted development to automate and scale business processes. Eager to bring strong attention to detail and cross-functional curiosity to the financial operations of a fast-growing, cross-border technology company.

Education

IIITDM Jabalpur

2023 – Present

B.Tech in Smart Manufacturing

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, C++

Data & Analysis: Pandas, NumPy, SciPy, Statistics & Hypothesis Testing, A/B Testing, Financial Impact Modeling, Excel / Google Sheets

BI & Reporting: Power BI (DAX), Metabase, Data Warehousing, Dashboarding

Software Tools: FastAPI, Docker, REST APIs, Git/GitHub, CI/CD (GitHub Actions), AI-Assisted Development

Databases: PostgreSQL, MongoDB, Supabase

Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design

Projects

Conformal-Demand-Forecasting

Python, LightGBM, FastAPI, Docker

Inventory Cost Optimization & Risk-Bounded Financial Planning

- Built a probabilistic demand-forecasting engine (multi-quantile LightGBM + Conformalized Quantile Regression) producing statistically guaranteed prediction intervals for inventory planning.
- Mapped calibrated intervals to optimal order quantities via Newsvendor cost optimization, cutting stockout rate from 48% to 11% and reducing total inventory holding cost by 31% across 3,680 test items — direct working-capital impact.
- Shipped the model as a containerized, auditable FastAPI microservice with automated testing and CI, built end-to-end with AI-assisted development for reliability.

OptiMetrics

Python, SciPy, Power BI, DAX

Statistical Experiment Audit & Financial Risk Reporting

- Audited a checkout experiment's headline +48% conversion lift and uncovered a hidden discrepancy: mobile conversions had actually collapsed by 47%, masked by aggregate reporting.
- Quantified the financial exposure of a full rollout and recommended a segmented deployment strategy, projected to avoid \$96,300 in lost revenue per 50K users.
- Built an executive Power BI reporting layer with custom DAX measures, translating raw statistical output into board-ready financial findings.

Olist-Analytics-Engine

PostgreSQL, SQL, Metabase, Docker

End-to-End Financial & Operational Reporting Pipeline

- Engineered a 4-layer SQL data warehouse (staging → intermediate → marts) over 100K+ real transaction records, using window functions and CTEs to build reusable reporting views.
- Quantified an estimated R\$1.73M revenue opportunity from logistics-delay and customer-segment analysis, packaged into a stakeholder-facing reporting layer.
- Containerized the full warehouse (Postgres, pgAdmin, Metabase) via Docker for reproducible, auditable reporting infrastructure.

Patent

Automatic Footrest Assembly for Two Wheeler

2025

- Designed a sensor-based automation system using embedded controllers and pressure sensors.
- Implemented hardware-software integration workflows with fail-safe operational logic.

Achievements

- **Google Cloud Arcade Facilitator Program** - deployed cloud-native applications on GCP using Compute Engine and Cloud Run.
- Selected for **Amazon ML Summer School 2026 (AMLSS)**, a competitive program focused on Machine Learning and Applied Data Science.

Co-Curricular Activities

- Organized and managed College Literature Club events, coordinating execution, logistics, and team workflows.
- Active member of a community Run Club, co-organizing running events and group games.